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UNIT-3

LECTURE-1

Public Key Cryptosystems with Applications:

A look at the encryption algorithm and its securitybenefits

Public-key cryptography, or asymmetric cryptography, is an encryption scheme that uses two mathematically related, but not identical, keys - a public key and a private key. Unlike symmetric key algorithms that rely on one key to both encrypt and decrypt, each key performs a unique function. The public key is used to encrypt and the private key is used to decrypt.

It is computationally infeasible to compute the private key based on the public key. Because of this, public keys can be freely shared, allowing users an easy and convenient method for encrypting content and verifying digital signatures, and private keys can be kept secret, ensuring only the owners of the private keys can decrypt content and create digital signatures.

Since public keys need to be shared but are too big to be easily remembered, they are stored on digital certificates for secure transport and sharing. Since private keys are not shared, they are simply stored in the software or operating system you use, or on hardware (e.g., USB token, hardware security module) containing drivers that allow it to be used with your software or operating system.

Digital certificates are issued by entities known as Certificate Authorities (CAs). For more information on CAs, please see our related article - What are Certificate Authorities?.

Business Applications

The main business applications for public-key cryptography are:

- **Digital signatures** - content is digitally signed with an individual's private key and is verified by the individual's public key
- **Encryption** - content is encrypted using an individual's public key and can only be decrypted with the individual's private key

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Security Benefits of Digital Signatures

Assuming the private key has remained secret and the individual it was issued to is the only person with access to it, digitally signing documents and emails offers the following benefits.

- **Authentication** – since the individual's unique private key was used to apply the signature, recipients can be confident that the individual was the one to actually apply the signature
- **Non-repudiation** – since the individual is the only one with access to the private key used to apply the signature, he/she cannot later claim that it wasn't him/her who applied the signature
- **Integrity** - when the signature is verified, it checks that the contents of the document or message match what was in there when the signature was applied. Even the slightest change to the original document would cause this check to fail.

Security Benefits of Encryption

Assuming the individual's private key has not been compromised, encrypting data and messages offers the following security benefits.

- **Confidentiality** - because the content is encrypted with an individual's public key, it can only be decrypted with the individual's private key, ensuring only the intended recipient can decrypt and view the contents/li>
- **Integrity** - part of the decryption process involves verifying that the contents of the original encrypted message and the new decrypted match, so even the slightest change to the original content would cause the decryption process to fail

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LECTURE-2

Requirements and Cryptanalysis:

Cryptanalyst

The primary responsibility of a cryptanalyst is to develop mathematical formulas and codes that protect data from prying eyes including cybercriminals and computer hackers. Public and private sector organizations hire cryptanalyst to ensure that any information/ data shared through their network is completely secure and encrypted. Computational number schemes and number theories are majorly used for writing codes to encrypt data or information. In simple words, cryptanalysts are expert mathematicians who are able to design, develop, and evaluation algorithms that can be used to decipher various number theory problems. Moreover, the responsibility of a cryptanalyst involves writing new encryption codes to secure data if a hacker or cybercriminal manages to decipher the existing codes. The responsibilities on the shoulder of a cryptanalyst are huge as their organization's image and financial stability depend on their expertise.

Generally, cryptanalysts have completed their undergraduate degrees in either mathematics or computer science and are said to possess excellent knowledge or math, economics, computer science, and English. However, to advance in the field and most cryptanalysts at senior positions have master's degree or Ph.D. in mathematics or computer science.

Individuals who are passionate about research and development activities and wish to advance their careers in the field can opt to become a university teacher, but this career choice would require them to have Ph.D. in the field.

What is Cryptanalysis?

You came here with the wish to know how to become a cryptanalyst, so it would be unfair if we don't dive into details about what cryptanalysis is in order to help you better understand what lies ahead – should you choose to become a cryptanalyst. It the science of decrypting coded data and turn it into plain text that is easy to understand. Cryptanalysis is usually the best way to transmit data over insecure networking channels. Cryptanalysis also entails encryption of data into codes so that unauthorized access to data can be prevented.

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Should you choose to become a cryptanalyst, you will be expected by your employer to write or decipher secret codes, which cybercriminals and hackers cannot crack. Cryptanalysts are professionals with the ability to analyze and decode secret messages. Whereas, they are also expected (in case if they are working for a government agency) to design coding systems that are immune to any potential threat. A cryptanalyst role in a private organization is to ensure the security of data and systems while protecting the privacy of the organization and its clients.

Differences in Cryptanalyst, Cryptographer or Cryptologist?

These terms can confuse a beginner as they all appear to focus on encryption and decryption of communication. But, let us explain it to you in a way that their difference becomes visible to you. Cryptology is the art and science of creating algorithms to protect confidential information/ data, whereas cryptanalysis deals with breaking the cryptographic algorithms. The mixture of these two terms is known as cryptology.

So what's the difference between a cryptographer, cryptanalyst, and cryptologists? This question may also pop up in your mind, thus to educate our readers about how to become a cryptanalyst, it is imperative to explain the differences of roles. A cryptographer is expected to design algorithms, develop encryption codes, and create impenetrable security systems so that confidential information that their employers wish to protect can be protected. Cryptologists are responsible for both i.e. creating codes and solving them. In layman terms, cryptography refers to the art and science of creating codes, whereas cryptanalysis is the art and science of breaking them and turning them into useful information.

Simply put, a cryptanalyst is required to perform analysis and decryption of the encrypted information. As a cryptanalyst, you may be required to decipher encrypted data in different cryptographic security systems.

Encrypted messages or data have two aspects one is internals and the other is externals. It is the responsibility of a cryptanalyst to identify the internals – a process of identifying how the message has been encrypted and this can be done by utilizing the externals of the message. A cryptanalyst is considered good if that person has the knowledge of applying analysis tools to identify externals, extract the key and subsequently decrypt the message. Once done with this process, the person should also be able to reconstruct the process of encryption applied to encrypt that message.

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To become a cryptanalyst, you should have adequate knowledge about ciphers, cryptosystems, ciphertexts and should develop the capability to identify and enhance techniques that can be considered as vulnerabilities of the system. A cryptanalyst has to decode a message or decrypt the data without having prior knowledge of the encryption key, plain text or the algorithm used during the process of encryption. Moreover, a cryptanalyst has to get the job done by targeting digital signatures, cryptographic algorithms, and secure hashing.

Definitions of Common Cryptanalysis Terms

Plaintext: This term in cryptography refers to a message in its original form – before encryption and after decryption.

Ciphertext: This term refers to a message that has been encrypted via a code. Ciphertext in simple words is the coded message or the result you get after encryption. However, cipher has been used interchangeably with the term ciphertext, which is inaccurate as it refers to the encryption process, not the encrypted message.

Hashing: It is another common term you would encounter if you wish to become a cryptanalyst, which basically refers to the process of changing of a string of characters into a fixed-length key, which becomes the representation of an actual string.

What does a Cryptanalyst do?

Having covered the basics of cryptanalysis, let's discuss what the responsibilities of cryptanalysts are. They are generally expected to develop mathematical models and write encryption codes to protect data from cybercriminals. They are also tasked to decipher codes and decrypt encrypted data for companies or government agencies. The job requirements of a cryptanalyst vary from employer to employer, but the basic responsibilities of encrypting and decrypting data remain the same. In a data-driven business environment, data protection has become one of the major concerns of government agencies and private corporations, hence they hire cryptanalysts to ensure that their networks, computer systems, and data remain safe and secure.

A cryptanalyst is needed by almost every company in every industry including retail, telecom, banks, e-commerce, government agencies, and others. Cryptanalyst can decide to develop expertise for any specific industry as well to further their

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career. For instance, in the banking industry, a cryptanalyst can develop expertise in securing the entire banking operations including automatic teller machines (ATMs), emails, credit cards, online banking, and consumer data. In the banking industry, cryptanalysts use ciphers to protect all the sensitive data including bank account numbers and credit card numbers. Cryptanalysts are also required by certain organizations to write codes and develop programs that secure communication channels and protect the network from any unauthorized access.

Computational number theories and number schemes are mainly used to write most of the encryption codes. Cryptanalysts are expected to have excellent command over mathematics since they have to solve a variety of problems using their expertise in mathematics.

Educational Requirements

The responsibilities mentioned above make it clear that the job of a cryptanalyst entails computer science and mathematics. Cryptanalysts mostly have a bachelor's degree in computer engineering, computer science or mathematics. However, some also have a master's degree in these fields as well.

When it comes to mathematics expertise required to become a cryptanalyst, the list is quite an extensive one and covers everything from linear algebra, elementary number theory, Calculus (I, II, & III), differential equations, mathematical cryptography, and methods of complex analysis among others.

Cryptanalysts have to rely on mathematics to fulfill a number of responsibilities including:

- Examine & test concepts as well as alternative theories
- Write encryption codes and encrypt data, databases, and systems
- Do cryptic computation and utilize numerical analysis methods
- Follow mathematical modules, formulas, and theorems

There are a number of responsibilities that cryptanalysts have to fulfill using computer engineering or computer science including:

- Design and develop robust computer systems that hackers cannot penetrate
- Recommend security protocols that the company should follow to protect its data
- Prepare data charts, tables, & graphs

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To be considered by an organization for the post of cryptanalyst, you have to have a degree in computer science, computer engineering, mathematics, or other related subjects. However, you can also become a cryptanalyst by doing some online certifications training available at QuickStart.

However, you don't need any technical degree or certification if you have a degree in mathematics, computer engineering or computer science along with necessary experience and training, since this is what employers seek when considering someone for the position of cryptanalyst. Moreover, there are companies and government agencies who would consider your profile as a cryptanalyst even if you don't have a technical or non-technical degree, since they only need skills, knowledge, and expertise. However, some organizations would prefer only graduates with over 5 years of experience to handle the privacy and security of their sensitive data. To be considered for an entry-level position, you may be required to demonstrate skills and expertise only along with a bachelor's or master's degree, needless to say, that you may have to compromise on your expected salary figures initially. If you wish to apply for a mid-level role, you may need to acquire at least 3 years of experience, otherwise, your profile won't be considered. Simply put, where you land a job depends on your hard work 90% of the times and luck has only 10% shares in it.

If you wish to pursue a degree in computer science or computer engineering, then your focus should be on the following subjects, should you choose to start your career as a cryptanalyst:

- C, Python, C++, Java or similar languages
- Computer Algorithms
- Computer Architecture
- Computer Data structures

To add attraction to your CV, QuickStart offers a number of information security courses & training that may further your knowledge about encryption, decryption, data security and others. Explore the courses available to see which skill you need the most. There are training and courses for everyone – so whether you are just a beginner or a professional, you'll find a course/ training that suits your needs. As a beginner, you may need to strengthen your knowledge about basic cryptographic algorithms, computational number theories, programming languages and more. If you successfully finish any of the course or training then the certificate you'll receive for it can be presented as a proof of acquired skills and may increase the probability of you getting your dream job. These specialized courses equip you

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with the knowledge you may need to work with a particular employer and may get the understanding to design and develop systems that are beneficial for the organization.

QuickStart is one of the leading platforms for getting reliable online certifications and knowledge to expand your area of expertise.

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LECTURE-3

RSA cryptosystem:

RSA Algorithm in Cryptography

RSA (**Rivest–Shamir–Adleman**) is a public-key cryptosystem that is widely used for secure data transmission. It is also one of the oldest. The acronym **RSA** comes from the surnames of Ron Rivest, Adi Shamir, and Leonard Adleman, who publicly described the algorithm in 1977. An equivalent system was developed secretly, in 1973 at GCHQ (the British signals intelligence agency), by the English mathematician Clifford Cocks. That system was declassified in 1997.[1]

In a public-key cryptosystem, the encryption key is public and distinct from the decryption key, which is kept secret (private). An RSA user creates and publishes a public key based on two large prime numbers, along with an auxiliary value. The prime numbers are kept secret. Messages can be encrypted by anyone, via the public key, but can only be decoded by someone who knows the prime numbers.[2]

The security of RSA relies on the practical difficulty of factoring the product of two large prime numbers, the "factoring problem". Breaking RSA encryption is known as the RSA problem. Whether it is as difficult as the factoring problem is an open question.[3] There are no published methods to defeat the system if a large enough key is used.

RSA is a relatively slow algorithm. Because of this, it is not commonly used to directly encrypt user data. More often, RSA is used to transmit shared keys for symmetric key cryptography, which are then used for bulk encryption-decryption.

RSA algorithm is asymmetric cryptography algorithm. Asymmetric actually means that it works on two different keys i.e. **Public Key** and **Private Key.** As the name describes that the Public Key is given to everyone and Private key is kept private.

An example of asymmetric cryptography :

1. A client (for example browser) sends its public key to the server and requests for some data.
2. The server encrypts the data using client's public key and sends the encrypted data.
3. Client receives this data and decrypts it.

Since this is asymmetric, nobody else except browser can decrypt the data even if a third party has public key of browser.

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The idea! The idea of RSA is based on the fact that it is difficult to factorize a large integer. The public key consists of two numbers where one number is multiplication of two large prime numbers. And private key is also derived from the same two prime numbers. So if somebody can factorize the large number, the private key is compromised. Therefore encryption strength totally lies on the key size and if we double or triple the key size, the strength of encryption increases exponentially. RSA keys can be typically 1024 or 2048 bits long, but experts believe that 1024 bit keys could be broken in the near future. But till now it seems to be an infeasible task.

Let us learn the mechanism behind RSA algorithm :

>> **Generating Public Key :**

- Select two prime no's. Suppose **P = 53 and Q = 59**.
- Now First part of the Public key : **n = P*Q = 3127**.
-
- We also need a small exponent say **e** :
- But e Must be
  -
    - An integer.
    -
    - Not be a factor of n.
    -
    - **1 < e <** Φ(n) [Φ(n) is discussed below],
    - Let us now consider it to be equal to 3.
-

- Our Public Key is made of n and e

>> **Generating Private Key :**

- We need to calculate Φ(n) :
- Such that **Φ(n) = (P-1)(Q-1)**
- so, Φ(n) = 3016
-
- Now calculate Private Key, **d** :
- **d = (k*Φ(n) + 1) / e** for some integer k
- For k = 2, value of d is 2011.

Now we are ready with our – Public Key (n = 3127 and e = 3) and Private Key(d = 2011)

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Now we will encrypt **“HI”** :

- Convert letters to numbers : H = 8 and I = 9
-
- Thus **Encrypted Data c = 89^e mod n**.
- Thus our Encrypted Data comes out to be 1394
-
- Now we will decrypt **1394** :
-
- **Decrypted Data = c^d mod n**.
- Thus our Encrypted Data comes out to be 89
- **8 = H and I = 9 i.e. "HI".**

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LECTURE-4

Rabin cryptosystem:

The **Rabin cryptosystem** is an asymmetric cryptographic technique, whose security, like that of RSA, is related to the difficulty of integer factorization. However the Rabin cryptosystem has the advantage that it has been mathematically proven to be computationally secure against a chosen-plaintext attack as long as the attacker cannot efficiently factor integers, while there is no such proof known for RSA. It has the disadvantage that each output of the Rabin function can be generated by any of four possible inputs; if each output is a ciphertext, extra complexity is required on decryption to identify which of the four possible inputs was the true plaintext.

Rabin Cryptosystem is an public-key cryptosystem invented by Michael Rabin. It uses asymmetric key encryption for communicating between two parties and encrypting the message.
The security of Rabin cryptosystem is related to the difficulty of factorization. It has the advantage over the others that the problem on which it banks has proved to be hard as **integer factorization**. It has the disadvantage also, that each output of the Rabin function can be generated by any of four possible inputs. if each output is a ciphertext, extra complexity is required on decryption to identify which of the four possible inputs was the true plaintext.

Steps in Rabin cryptosystem

Key generation

Generate two very large prime numbers, p and q, which satisfies the condition
$p \neq q \rightarrow p \equiv q \equiv 3 \pmod 4$
For example:
p=139 and q=191
Calculate the value of n
n = p.q
Publish n as public key and save p and q as private key

Encryption

1. Get the public key n.
2. Convert the message to ASCII value. Then convert it to binary and extend the binary value with itself, and change the binary value back to decimal m.

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Encrypt with the formula:
$C = m^2 \bmod n$
Send C to recipient.

Decryption

Accept C from sender.
Specify a and b with Extended Euclidean GCD such that, a.p + b.q = 1
Compute r and s using following formula:
$r = C^{(p+1)/4} \bmod p$
$s = C^{(q+1)/4} \bmod q$
Now, calculate X and Y using following formula:
$X = (a.p.r + b.q.s) \bmod p$
$Y = (a.p.r – b.q.s) \bmod q$
The four roots are, m1=X, m2=-X, m3=Y, m4=-Y
Now, Convert them to binary and divide them all in half.

1. Determine in which the left and right half are same. Keep that binary's one half and convert it to decimal m. Get the ASCII character for the decimal value m. The resultant character gives the correct message sent by sender.

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LECTURE-5

Elgamal cryptosystem:

"ElGamal" redirects here. For signature algorithm, see ElGamal signature scheme.

In cryptography, the **ElGamal encryption system** is an asymmetric key encryption algorithm for public-key cryptography which is based on the Diffie–Hellman key exchange. It was described by Taher Elgamal in 1985. ElGamal encryption is used in the free GNU Privacy Guard software, recent versions of PGP, and other cryptosystems. The Digital Signature Algorithm (DSA) is a variant of the ElGamal signature scheme, which should not be confused with ElGamal encryption.

ElGamal encryption can be defined over any cyclic group , like multiplicative group of integers modulo *n*. Its security depends upon the difficulty of a certain

problem in related to computing discrete logarithms.

ElGamal encryption is an public-key cryptosystem. It uses asymmetric key encryption for communicating between two parties and encrypting the message. This cryptosystem is based on the difficulty of finding **discrete logarithm** in a cyclic group that is even if we know g^a and g^k, it is extremely difficult to compute g^{ak}.

Idea of ElGamal cryptosystem

Suppose Alice wants to communicate to Bob.

1. Bob generates public and private key :
   - Bob chooses a very large number **q** and a cyclic group $\mathbf{F}_q$.
   - From the cyclic group $\mathbf{F}_q$, he choose any element **g** and an element **a** such that gcd(a, q) = 1.
   - Then he computes $h = g^a$.
   - Bob publishes **F**, $\mathbf{h = g^a}$, **q** and **g** as his public key and retains **a** as private key.
2. Alice encrypts data using Bob's public key :
   - Alice selects an element **k** from cyclic group **F** such that gcd(k, q) = 1.
   - Then she computes $p = g^k$ and $s = h^k = g^{ak}$.
   - She multiples s with M.

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- Then she sends (p, M*s) = (g^k, M*s).

3. Bob decrypts the message :
   - Bob calculates $s' = p^a = g^{ak}$.
   - He divides M*s by s′ to obtain M as s = s′.

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LECTURE-6

Elliptic curve:

Elliptic curves are curves defined by a certain type of cubic equation in two variables. The set of rational solutions to this equation has an extremely interesting structure, including a group law. The theory of elliptic curves was essential in Andrew Wiles' proof of Fermat's last theorem. Computational problems involving the group law are also used in many cryptographic applications, and in algorithms for factoring large integers.

ECC, an alternative technique to RSA, is a powerful cryptography approach. It generates security between key pairs for public key encryption by using the mathematics of elliptic curves.

RSA does something similar with prime numbers instead of elliptic curves, but ECC has gradually been growing in popularity recently due to its smaller key size and ability to maintain security. This trend will probably continue as the demand on devices to remain secure increases due to the size of keys growing, drawing on scarce mobile resources. This is why it is so important to understand elliptic curve cryptography in context.

In contrast to RSA, ECC bases its approach to public key cryptographic systems on how elliptic curves are structured algebraically over finite fields. Therefore, ECC creates keys that are more difficult, mathematically, to crack. For this reason, ECC is considered to be the next generation implementation of public key cryptography and more secure than RSA.

It also makes sense to adopt ECC to maintain high levels of both performance and security. That's because ECC is increasingly in wider use as websites strive for greater online security in customer data and greater mobile optimization, simultaneously. More sites using ECC to secure data means a greater need for this kind of quick guide to elliptic curve cryptography.

An elliptic curve for current ECC purposes is a plane curve over a finite field which is made up of the points satisfying the equation: $y^2=x^3 + ax + b$.

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In this elliptic curve cryptography example, any point on the curve can be mirrored over the x-axis and the curve will stay the same. Any non-vertical line will intersect the curve in three places or fewer.

Elliptic Curve Cryptography vs RSA

The difference in size to security yield between RSA and ECC encryption keys is notable. The table below shows the sizes of keys needed to provide the same level of security. In other words, an elliptic curve cryptography key of 384 bit achieves the same level of security as an RSA of 7680 bit.

RSA Key Length (bit)
1024
2048
3072
7680
15360

ECC Key Length (bit)
160
224
256
384
521

There is no linear relationship between the sizes of ECC keys and RSA keys. That is, an RSA key size that is twice as big does not translate into an ECC key size that's doubled. This compelling difference shows that ECC key generation and signing are substantially quicker than for RSA, and also that ECC uses less memory than does RSA.

Also, unlike in RSA, where both are integers, in ECC the private and public keys are not equally exchangeable. Instead, in ECC the public key is a point on the curve, while the private key is still an integer.

A quick comparison of the advantages and disadvantages of ECC and RSA algorithms looks like this:

ECC features smaller ciphertexts, keys, and signatures, and faster generation of keys and signatures. Its decryption and encryption speeds are moderately fast. ECC enables lower latency than inverse throughout by computing signatures in two

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stages. ECC features strong protocols for authenticated key exchange and support for the tech is strong.

The main disadvantage of ECC is that it isn't easy to securely implement. Compared to RSA, which is much simpler on both the verification and encryption sides, ECC is a steeper learning curve and a bit slower for accumulating actionable results.

However, the disadvantages of RSA catch up with you soon. Key generation is slow with RSA, and so is decryption and signing, which aren't always that easy to implement securely.

Advantages of Elliptic Curve Cryptography

Public-key cryptography works using algorithms that are easy to process in one direction and difficult to process in the reverse direction. For example, RSA relies on the fact that multiplying prime numbers to get a larger number is easy, while factoring huge numbers back to the original primes is much more difficult.

However, to remain secure, RSA needs keys that are 2048 bits or longer. This makes the process slow, and it also means that key size is important.

Size is a serious advantage of elliptic curve cryptography, because it translates into more power for smaller, mobile devices. It's far simpler and requires less energy to factor than it is to solve for an elliptic curve discrete logarithm, so for two keys of the same size, RSA's factoring encryption is more vulnerable.

Using ECC, you can achieve the same security level using smaller keys. In a world where mobile devices must do more and more cryptography with less computational power, ECC offers high security with faster, shorter keys compared to RSA.

How Secure is Elliptic Curve Cryptography?

There are several potential vulnerabilities to elliptic curve cryptography, including side-channel attacks and twist-security attacks. Both types aim to invalidate the ECC's security for private keys.

Side-channel attacks including differential power attacks, fault analysis, simple power attacks, and simple timing attacks, typically result in information leaks. Simple countermeasures exist for all types of side-channel attacks.

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An additional type of elliptic curve attack is the twist-security attack or fault attack. Such attacks may include invalid-curve attacks and small-subgroup attacks, and they may result in the private key of the victim leaking out. Twist-security attacks are typically simply mitigated with careful parameter validation and curve choices.

Although there are certain ways to attack ECC, the advantages of elliptic curve cryptography for wireless security mean it remains a more secure option.

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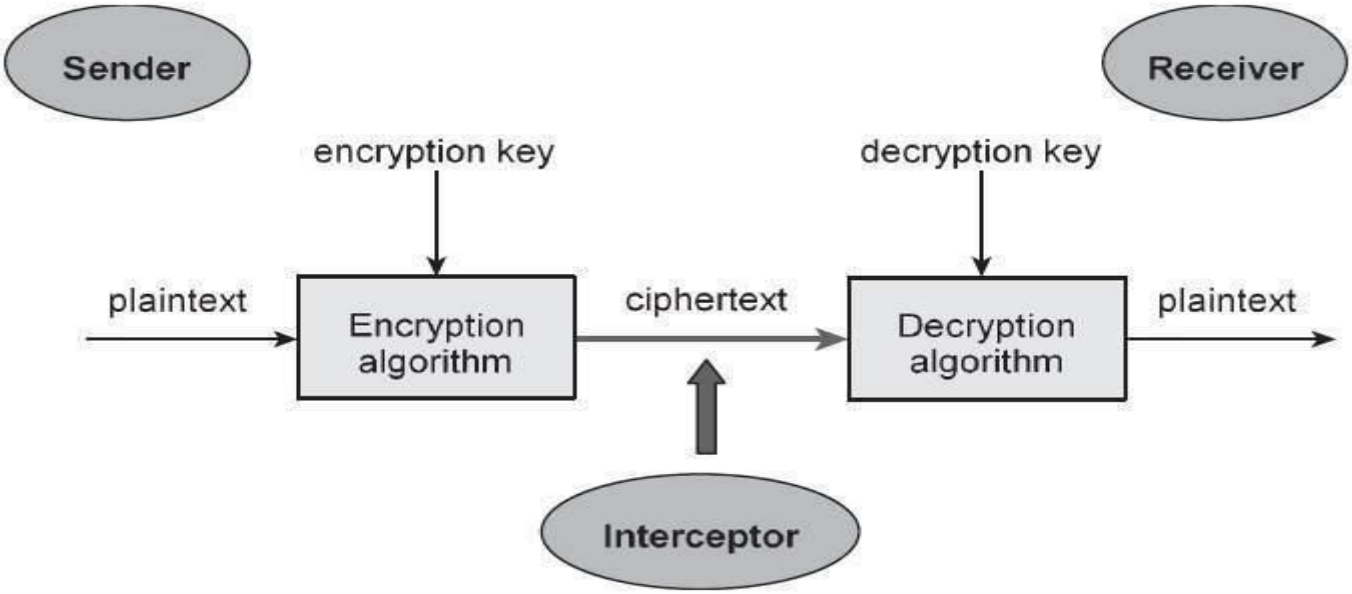
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LECTURE-7

Cryptosystem:

A cryptosystem is an implementation of cryptographic techniques and their accompanying infrastructure to provide information security services. A cryptosystem is also referred to as a **cipher system**.

Let us discuss a simple model of a cryptosystem that provides confidentiality to the information being transmitted. This basic model is depicted in the illustration below −



The illustration shows a sender who wants to transfer some sensitive data to a receiver in such a way that any party intercepting or eavesdropping on the communication channel cannot extract the data.

The objective of this simple cryptosystem is that at the end of the process, only the sender and the receiver will know the plaintext.

Components of a Cryptosystem

The various components of a basic cryptosystem are as follows −

- **Plaintext.** It is the data to be protected during transmission.
- **Encryption Algorithm.** It is a mathematical process that produces a ciphertext for any given plaintext and encryption key. It is a cryptographic

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algorithm that takes plaintext and an encryption key as input and produces a ciphertext.

- **Ciphertext.** It is the scrambled version of the plaintext produced by the encryption algorithm using a specific the encryption key. The ciphertext is not guarded. It flows on public channel. It can be intercepted or compromised by anyone who has access to the communication channel.
- **Decryption Algorithm,** It is a mathematical process, that produces a unique plaintext for any given ciphertext and decryption key. It is a cryptographic algorithm that takes a ciphertext and a decryption key as input, and outputs a plaintext. The decryption algorithm essentially reverses the encryption algorithm and is thus closely related to it.
- **Encryption Key.** It is a value that is known to the sender. The sender inputs the encryption key into the encryption algorithm along with the plaintext in order to compute the ciphertext.
- **Decryption Key.** It is a value that is known to the receiver. The decryption key is related to the encryption key, but is not always identical to it. The receiver inputs the decryption key into the decryption algorithm along with the ciphertext in order to compute the plaintext.

For a given cryptosystem, a collection of all possible decryption keys is called a **key space**.

An **interceptor** (an attacker) is an unauthorized entity who attempts to determine the plaintext. He can see the ciphertext and may know the decryption algorithm. He, however, must never know the decryption key.

Types of Cryptosystems

Fundamentally, there are two types of cryptosystems based on the manner in which encryption-decryption is carried out in the system −

- Symmetric Key Encryption
- Asymmetric Key Encryption

The main difference between these cryptosystems is the relationship between the encryption and the decryption key. Logically, in any cryptosystem, both the keys are closely associated. It is practically impossible to decrypt the ciphertext with the key that is unrelated to the encryption key.

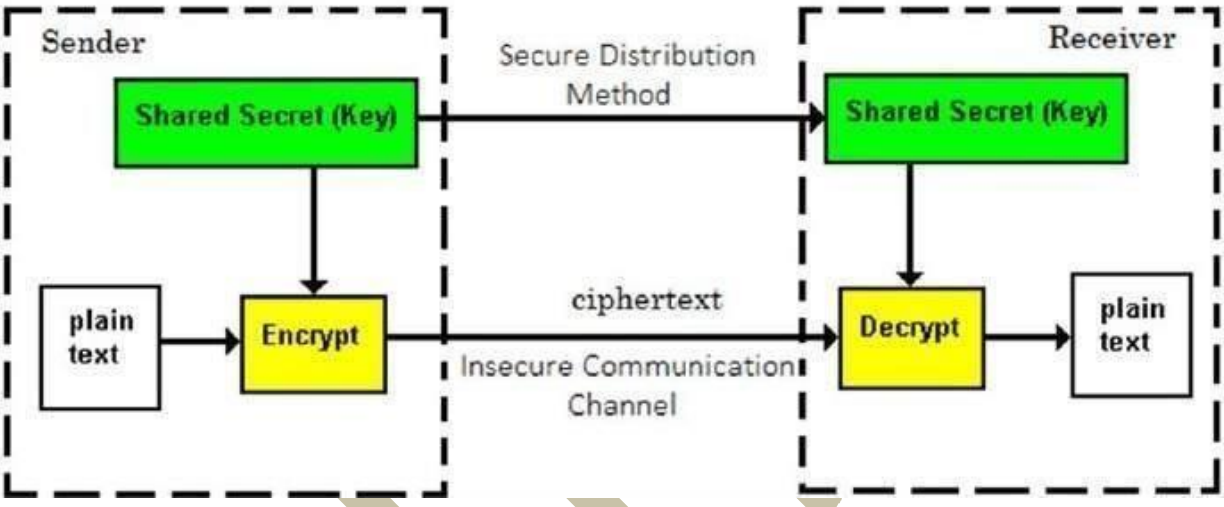
Symmetric Key Encryption

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The encryption process where **same keys are used for encrypting and decrypting** the information is known as Symmetric Key Encryption.

The study of symmetric cryptosystems is referred to as **symmetric cryptography**. Symmetric cryptosystems are also sometimes referred to as **secret key cryptosystems**.

A few well-known examples of symmetric key encryption methods are − Digital Encryption Standard (DES), Triple-DES (3DES), IDEA, and BLOWFISH.



Prior to 1970, all cryptosystems employed symmetric key encryption. Even today, its relevance is very high and it is being used extensively in many cryptosystems. It is very unlikely that this encryption will fade away, as it has certain advantages over asymmetric key encryption.

The salient features of cryptosystem based on symmetric key encryption are −

- Persons using symmetric key encryption must share a common key prior to exchange of information.
- Keys are recommended to be changed regularly to prevent any attack on the system.
- A robust mechanism needs to exist to exchange the key between the communicating parties. As keys are required to be changed regularly, this mechanism becomes expensive and cumbersome.
- In a group of **n** people, to enable two-party communication between any two persons, the number of keys required for group is **n × (n – 1)/2**.
- Length of Key (number of bits) in this encryption is smaller and hence, process of encryption-decryption is faster than asymmetric key encryption.

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- Processing power of computer system required to run symmetric algorithm is less.

Challenge of Symmetric Key Cryptosystem

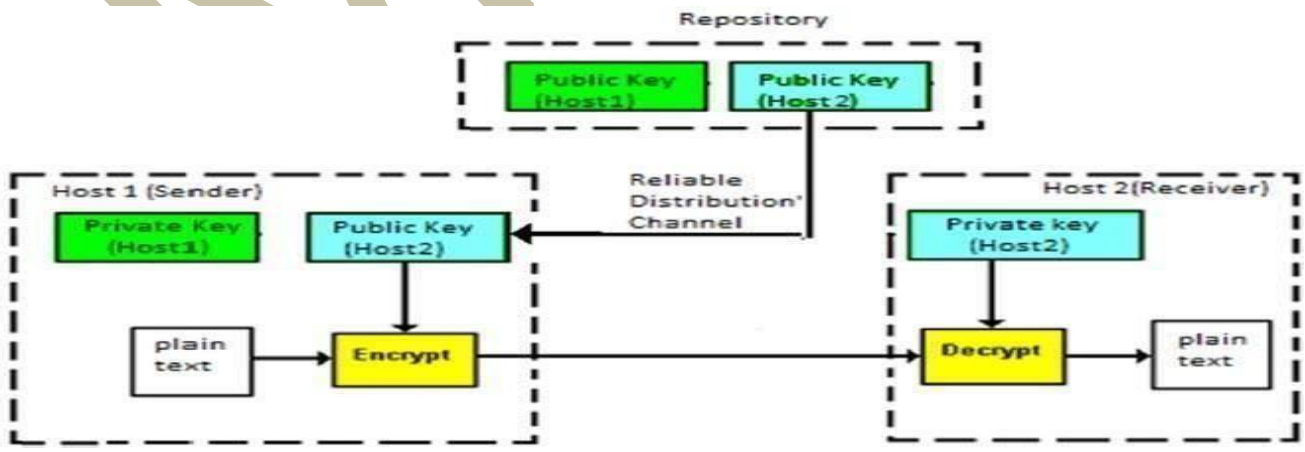
There are two restrictive challenges of employing symmetric key cryptography.

- **Key establishment** − Before any communication, both the sender and the receiver need to agree on a secret symmetric key. It requires a secure key establishment mechanism in place.
- **Trust Issue** − Since the sender and the receiver use the same symmetric key, there is an implicit requirement that the sender and the receiver 'trust' each other. For example, it may happen that the receiver has lost the key to an attacker and the sender is not informed.

These two challenges are highly restraining for modern day communication. Today, people need to exchange information with non-familiar and non-trusted parties. For example, a communication between online seller and customer. These limitations of symmetric key encryption gave rise to asymmetric key encryption schemes.

Asymmetric Key Encryption

The encryption process where **different keys are used for encrypting and decrypting the information** is known as Asymmetric Key Encryption. Though the keys are different, they are mathematically related and hence, retrieving the plaintext by decrypting ciphertext is feasible. The process is depicted in the following illustration −



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Asymmetric Key Encryption was invented in the 20th century to come over the necessity of pre-shared secret key between communicating persons. The salient features of this encryption scheme are as follows −

- Every user in this system needs to have a pair of dissimilar keys, **private key** and **public key**. These keys are mathematically related − when one key is used for encryption, the other can decrypt the ciphertext back to the original plaintext.
- It requires to put the public key in public repository and the private key as a well-guarded secret. Hence, this scheme of encryption is also called **Public Key Encryption**.
- Though public and private keys of the user are related, it is computationally not feasible to find one from another. This is a strength of this scheme.
- When *Host1* needs to send data to *Host2*, he obtains the public key of *Host2* from repository, encrypts the data, and transmits.
- *Host2* uses his private key to extract the plaintext.
- Length of Keys (number of bits) in this encryption is large and hence, the process of encryption-decryption is slower than symmetric key encryption.
- Processing power of computer system required to run asymmetric algorithm is higher.

Symmetric cryptosystems are a natural concept. In contrast, public-key cryptosystems are quite difficult to comprehend.

You may think, *how can the encryption key and the decryption key are 'related', and yet it is impossible to determine the decryption key from the encryption key?* The answer lies in the mathematical concepts. It is possible to design a cryptosystem whose keys have this property. The concept of public-key cryptography is relatively new. There are fewer public-key algorithms known than symmetric algorithms.

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